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COMPUTER ARCHITECTURE AND ORGANIZATION

Computer Architecture & Organization: Hardware Design

Instruction Set Architectures, ALU Design, Pipelining, Cache Memory Hierarchies
& Multi-Core Systems

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CHAPTER 1

Instruction Set Architecture (ISA) & Data Representation

1. Theoretical Foundations & Scope

Computer architecture defines the programmer-visible interface to hardware. This chapter studies RISC versus CISC philosophies, fixed vs variable length instruction formats, endianness, and IEEE 754 floating-point standards.

2. Core Conceptual Framework

- **RISC vs CISC Philosophies:** Simple single-cycle instructions with load-store architecture versus complex multi-cycle microcoded instructions.
- **Instruction Cycle:** Fetch, decode, operand fetch, execute, memory access, and write-back register update sequences.
- **IEEE 754 Floating-Point:** Sign bit, biased exponent, and normalized fraction representation for 32-bit single precision floats.

Key Practical Insights & Examination Takeaways:

- ' Analyze trade-offs between hardware simplicity (RISC) and code density (CISC).
- ' Encode and decode assembly instructions into machine binary formats.
- ' Convert real numbers into IEEE 754 floating-point binary bit patterns.

CHAPTER 2

Processor Pipelining, Hazards & Branch Prediction

1. Theoretical Foundations & Scope

Pipelining increases processor instruction throughput by overlapping instruction execution stages. We analyze structural hazards, data hazards (RAW, WAR, WAW), control hazards, forwarding paths, and branch predictors.

2. Core Conceptual Framework

- **Pipelined Throughput:** Achieving ideal CPI (Cycles Per Instruction) approaching 1 through classic 5-stage RISC pipelines.
- **Data Hazards & Forwarding:** Bypassing pipeline register stalls by routing ALU results directly to waiting pipeline stages.
- **Control Hazards & Branch Prediction:** Minimizing branch penalties using 2-bit saturating counter predictors and Branch Target Buffers (BTB).

Key Practical Insights & Examination Takeaways:

- ' Calculate speedup ratios of pipelined processors versus non-pipelined hardware.
- ' Identify Read-After-Write (RAW) data dependencies in assembly streams.
- ' Explain why out-of-order execution engines require reorder buffers (ROB).

CHAPTER 3

Memory Hierarchies: Cache Mapping & Virtual Memory

1. Theoretical Foundations & Scope

The memory hierarchy bridges the latency gap between fast CPU registers and slower main memory. We examine temporal and spatial locality, cache placement policies (direct, set-associative, fully associative), write policies, and virtual memory page tables.

2. Core Conceptual Framework

- **Cache Mapping Schemes:** Tag, index, and offset address breakdown for direct mapped and N-way set associative caches.
- **Cache Miss Classification:** The 3 Cs: Compulsory (cold), Capacity, and Conflict misses, and replacement strategies like LRU.
- **Virtual Memory & TLBs:** Translating virtual addresses to physical frame addresses using multi-level page tables and Translation Lookaside Buffers.

Key Practical Insights & Examination Takeaways:

- ' Calculate cache hit/miss penalties and average memory access times (AMAT).
- ' Partition memory addresses into Tag, Index, and Offset fields.
- ' Explain how virtual memory enables process memory isolation and protection.

Academic References & Further Study

[1] Quality Education Academic Standards & Pedagogical Guidelines, 2026 Edition.

[2] International Computer Science & Engineering Curriculum Framework.

[3] Educational Technology & Assessment Methodology Research Series.

[4] Open Access Educational Resource Repository - Quality Education Platform.

